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10 CFR 50.73

GNRO2025-00002

January 9, 2025

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2024-005-00,
Loss of Seal Steam Controller Power Resulted in Manual Reactor
Shutdown

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report (LER) 2024-005-00, Loss of Seal Steam Controller Power Resulted in Manual Reactor Shutdown. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(iv)(A), for events or conditions that resulted in the manual or automatic actuation of the reactor protection system.

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact me at 802-380-5124.

Sincerely,

A handwritten signature in blue ink, appearing to read 'JAH', with a stylized flourish at the end.

JAH/saw

Attachments: Licensee Event Report 2024-005-00

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U.S Nuclear Regulatory Commission
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Washington, DC 20555-0001

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Attachment
Licensee Event Report 2024-005-00

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

1. Facility Name Grand Gulf Nuclear Station, Unit 1										☒ 050 ☐ 052		2. Docket Number 00416		3. Page 1 of 2	
4. Title Loss of Seal Steam Controller Power Resulted in Manual Reactor Shutdown															
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved						
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name N/A			☐ 050	Docket Number		
11	10	2024	2024	005	00	01	09	2025	Facility Name N/A			☐ 052	Docket Number		
9. Operating Mode 1									10. Power Level 100						
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)															
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)					
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)					
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)					
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)					
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)					
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)(i)		☐ 73.1200(f)					
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)					
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)					
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)									
☐ Other (Specify here, in abstract, or in NRC 366A).															
12. Licensee Contact for this LER															
Licensee Contact Jeff Hardy, Regulatory Assurance Manager										Phone Number (Include Area Code) 802-380-5124					
13. Complete One Line for each Component Failure Described in this Report															
Cause	System	Component	Manufacturer	Reportable To IRIS		Cause	System	Component	Manufacturer	Reportable To IRIS					
X	SJ	JX	G080	N											
14. Supplemental Report Expected										15. Expected Submission Date		Month	Day	Year	
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)														
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)															
On November 10, 2024, at 0337 CST, Grand Gulf Nuclear Station, Unit 1 was operating at 100 percent power with all safety systems operable. A failed power supply resulted in a loss of seal steam controller power and led to a degraded condenser vacuum which resulted in a manual reactor shutdown. All control rods were fully inserted with no complications. Following initial scram response, a group 1 containment isolation signal resulted in the closure of all main steam isolation valves at 0545 CST. Later, an automatic Reactor Protection System actuation occurred at 0620 CST while in MODE 3 while all control rods were inserted. The cause of the loss of seal steam controller power was determined to be a failed power supply, 1N21K602B which had exceeded its expected life. The 1N21K602B power supply was replaced, and seal steam control was returned to service. This report is made in accordance with 10 CFR 50.73(a)(2)(iv)(A) for events or conditions that resulted in the manual or automatic actuation of the reactor protection system.															

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Grand Gulf Nuclear Station, Unit 1	☒ 050	00416	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2024	005	00

NARRATIVE**Plant Conditions:**

At the time of the event, Grand Gulf Nuclear Station, Unit 1 was in Mode 1 at 100% power, and all safety systems were operable.

Description of Event:

On November 10, 2024, at 0337 CST, a power supply failure resulted in a loss of seal steam controller power and led to a degraded condenser vacuum which resulted in a manual reactor shutdown. All control rods fully inserted, and there were no complications.

Following initial scram response, a group 1 containment isolation signal resulted in the closure of all main steam isolation valves at 0545 CST. This signal was due to the continued degradation of the condenser vacuum post-trip. An automatic RPS actuation occurred while in MODE 3 with all control rods currently inserted at 0620 CST.

Following the second actuation of the reactor protection system (RPS), pressure was controlled by safety relief valves and reactor core isolation cooling was used to maintain water level.

Reportability:

This report is made in accordance with 10 CFR 50.73(a)(2)(iv)(A) for events or conditions which resulted in the manual or automatic actuation of the RPS.

Cause(s):

The direct cause of the 1N21K602B power supply failure was aged components exceeding their expected life.

The root cause of the event was the determination of the life expectancy of the power supplies with electrolytic capacitors was incorrectly designed using the in-service date as the starting criteria; however, electrolytic capacitors begin degrading upon manufactured date. Starting the life expectancy for this power supply from the in-service date as opposed to manufactured date resulted in the power supply being installed with a projected life expectancy of 10 years when the component already had 6 years of shelf life and only 4 years of remaining life expectancy.

Corrective Actions:

Completed Actions:

-The seal steam power supply 1N21K602B was replaced, and seal steam control was restored.

Planned Actions:

-Revise maintenance procedure to include steps which guide actual full age determination and evaluation of power supplies. This will include a signature requirement for Maintenance to contact Supply Chain, and it will require initiating a Condition Report to document and evaluate the power supply age prior to being installed.

Safety Significance:

There were no actual safety consequences associated with this condition. Plant systems responded as designed and the unit was brought to Mode 3 by Control Room Operators.

Previous Similar Events:

None identified.